Project: Creditworthiness

Complete each section. When you are ready, save your file as a PDF document and submit it here:
https://classroom.udacity.com/nanodegrees/nd008/parts/11a7bf4c-2b69-47f3-9aec-108ce847f855/project

Step 1: Business and Data Understanding

Provide an explanation of the key decisions that need to be made. (250 word limit)

Key Decisions:

Answer these questions

- What decisions needs to be made?

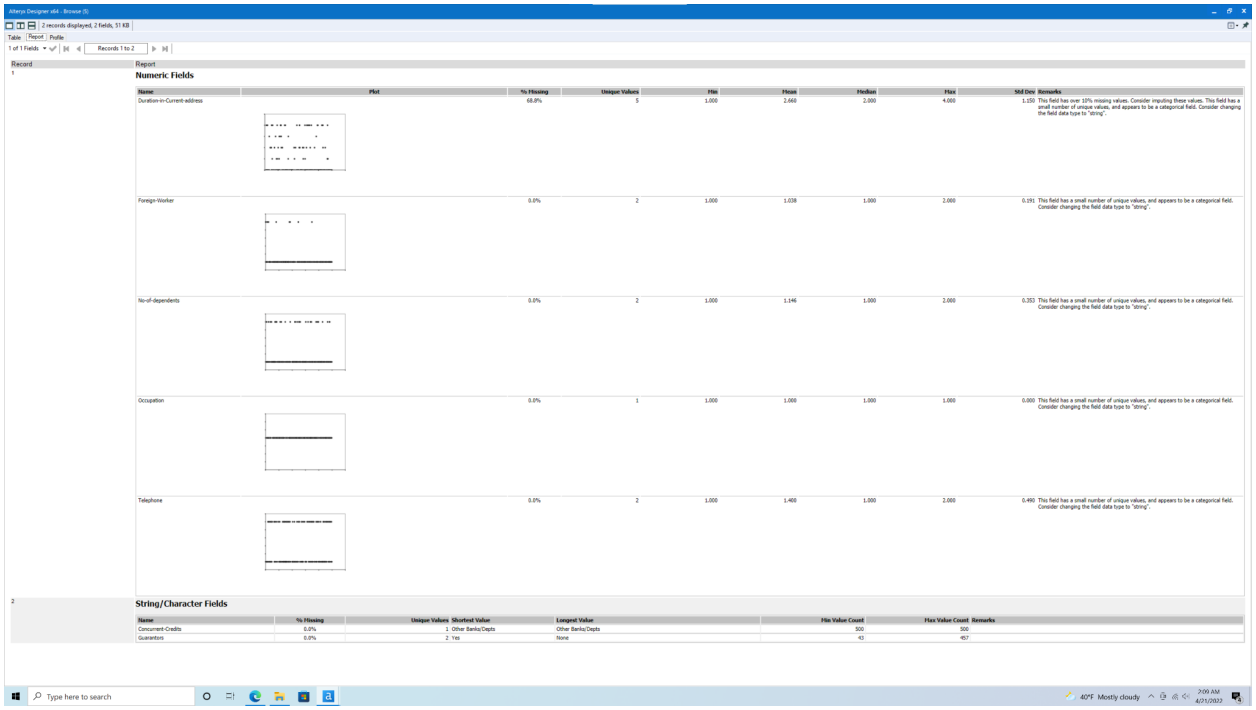
We need to determine whether the new customers based in the data provided are worthy for a loan from our bank.

- What data is needed to inform those decisions?
    - Account Balance
    - Duration of Credit
    - Credit Amount
    - Value Stocks
    - Purpose
    - Age
    - Type OF Apartment
    - Number of Credits at bank
    - Length of Employment
    - Installment Percent
    - Most Valualable Asset
    - PAyment Status of Last Credit

The data comes from two sources. 1 being Credit Training File and the list of customers that need to be processed in the next few days (Customers to Score File)

- What kind of model (Continuous, Binary, Non-Binary, Time-Series) do we need to use to help make these decisions?
    - The model will be binary on whether the credit is approved or not approved

Step 2: Building the Training Set



Numeric Fields

Name	Plot	% Missing	Unique Values	Min	Mean	Median	Max	Std Dev	Remarks
Duration-in-Current-address		68.0%	5	1.000	2.660	2.000	4.000	1.159	This field has over 10% missing values. Consider imputing these values. This field has a small number of unique values, and appears to be a categorical field. Consider changing the field data type to "string".
Foreign-Worker		0.0%	2	1.000	1.038	1.000	2.000	0.191	This field has a small number of unique values, and appears to be a categorical field. Consider changing the field data type to "string".
No-of-dependents		0.0%	2	1.000	1.146	1.000	2.000	0.353	This field has a small number of unique values, and appears to be a categorical field. Consider changing the field data type to "string".
Occupation		0.0%	1	1.000	1.000	1.000	1.000	0.000	This field has a small number of unique values, and appears to be a categorical field. Consider changing the field data type to "string".
Telephone		0.0%	2	1.000	1.400	1.000	2.000	0.490	This field has a small number of unique values, and appears to be a categorical field. Consider changing the field data type to "string".

String/Character Fields

Name	% Missing	Unique Values	Shortest Value	Longest Value	Min Value Count	Max Value Count	Remarks
Concurrent-Credits	0.0%	1	Other Banks/Depts	Other Banks/Depts	500	500	
Guarantors	0.0%	2	Yes	None	43	457	

I removed the fields of Foreign worker, Number of dependents, Occupation, Concurrent Credits, Guarantors, Duration in Address, and Telephone. These were chosen because they either had only two variables, empty records, or low variability. I then imputed the Age Years

Step 3: Train your Classification Models

Logistic Regression

The table below shows the variables where the p values are less than .05: Account.BalanceSomeBalance, Payment.Status.Of.Previous.CreditSome Problems, PurposeNew car, Credit.Amount, Length.of.current.employment< 1 year, Instalment.per.cent

- Which predictor variables are significant or the most important? Please show the p-values or variable importance charts for all of your predictor variables.
  Logistic Regression
    - The table below shows the variables where the p values are less than .05:
    - Account.BalanceSomeBalance, Payment.Status.Of.Previous.CreditSome

  Problems, PurposeNew car, Credit.Amount, Length.of.current.employment< 1 year, Instalment.per.cent

Record Report

1 **Report for Logistic Regression Model Stepwise_CreditWorthy**

2 *Basic Summary*

3 Call:
glm(formula = Credit.Application.Result ~ Account.Balance + Payment.Status.of.Previous.Credit + Purpose + Credit.Amount + Length.of.current.employment + Instalment.per.cent + Most.valuable.available.asset, family = binomial(logit), data = the.data)

4 Deviance Residuals:

5

Min	1Q	Median	3Q	Max
-2.289	-0.713	-0.448	0.722	2.454

6 Coefficients:

7

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-2.9621914	6.837e-01	-4.3326	1e-05 ***
Account.BalanceSome Balance	-1.6053228	3.067e-01	-5.2344	1.65e-07 ***
Payment.Status.of.Previous.CreditPaid Up	0.2360857	2.977e-01	0.7930	0.42775
Payment.Status.of.Previous.CreditSome Problems	1.2154514	5.151e-01	2.3595	0.0183 *
PurposeNew car	-1.6993164	6.142e-01	-2.7668	0.00566 **
PurposeOther	-0.3257637	8.179e-01	-0.3983	0.69042
PurposeUsed car	-0.7645820	4.004e-01	-1.9096	0.05618 .
Credit.Amount	0.0001704	5.733e-05	2.9716	0.00296 **
Length.of.current.employment4-7 yrs	0.3127022	4.587e-01	0.6817	0.49545
Length.of.current.employment< 1yr	0.8125785	3.874e-01	2.0973	0.03596 *
Instalment.per.cent	0.3016731	1.350e-01	2.2340	0.02549 *
Most.valuable.available.asset	0.2650267	1.425e-01	1.8599	0.06289 .

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for binomial taken to be 1)

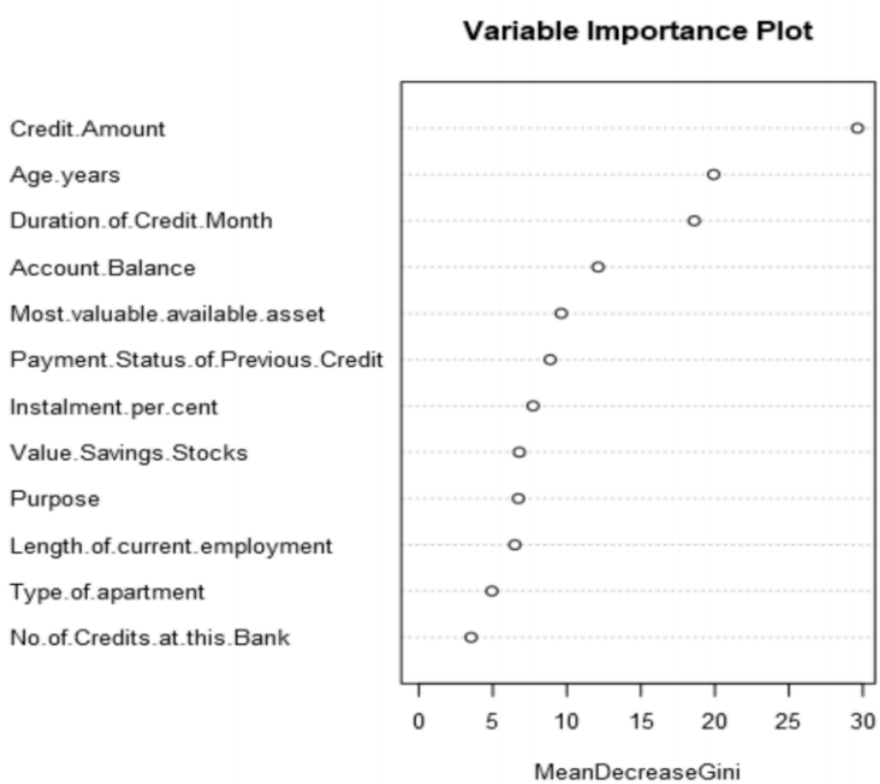
8 Null deviance: 413.16 on 349 degrees of freedom
Residual deviance: 328.55 on 338 degrees of freedom
McFadden R-Squared: 0.2048, AIC: 352.5

Forest Model

In this model the most important variables are Credit amount and Age

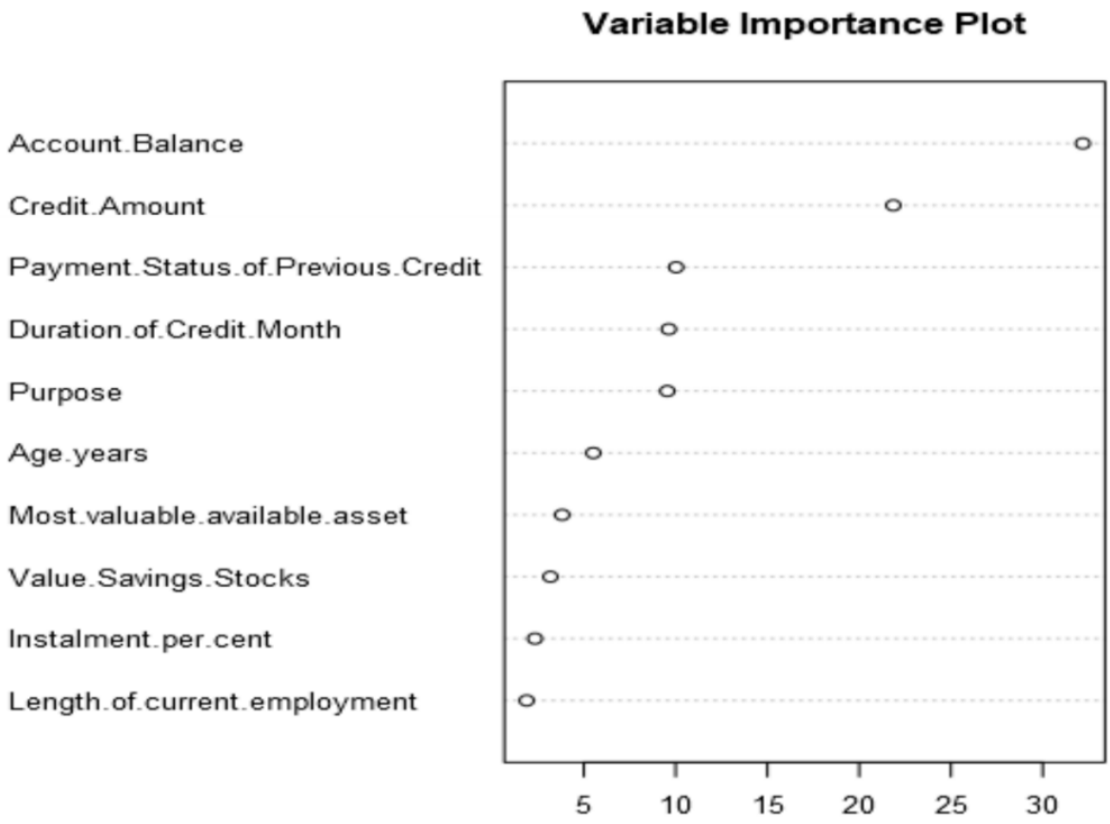
Report Profile

1 of 1 Fields Records 1 to 9



Boosted Model
The most important variables account balance and credit amount

Plots:



Decision Tree

Credit Application Result is the target variable while Account Balance, Duration and Credit are the most significant.

- Validate your model against the Validation set. What was the overall percent accuracy? Show the confusion matrix. Are there any bias seen in the model's predictions?

I did the model comparison report and the accuracy is shown:
Stepwise:0.7600
Decision Tree:0.6667
Boosted:0.7867
Forest:0.8067
There are bias in all of the models.

Model Comparison Report

Fit and error measures

Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Stepwise_Model	0.7600	0.8364	0.7306	0.8000	0.6286
Decision_Tree_Model	0.6667	0.7685	0.6272	0.7477	0.4359
Boosted_Model	0.7867	0.8632	0.7524	0.7829	0.8095
Forest_Model	0.8067	0.8755	0.7392	0.7969	0.8636

Confusion matrix of Boosted_Model

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	101	28
Predicted_Non-Creditworthy	4	17

Confusion matrix of Decision_Tree_Model

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	83	28
Predicted_Non-Creditworthy	22	17

Confusion matrix of Forest_Model

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	102	26
Predicted_Non-Creditworthy	3	19

Confusion matrix of Stepwise_Model

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	92	23
Predicted_Non-Creditworthy	13	22

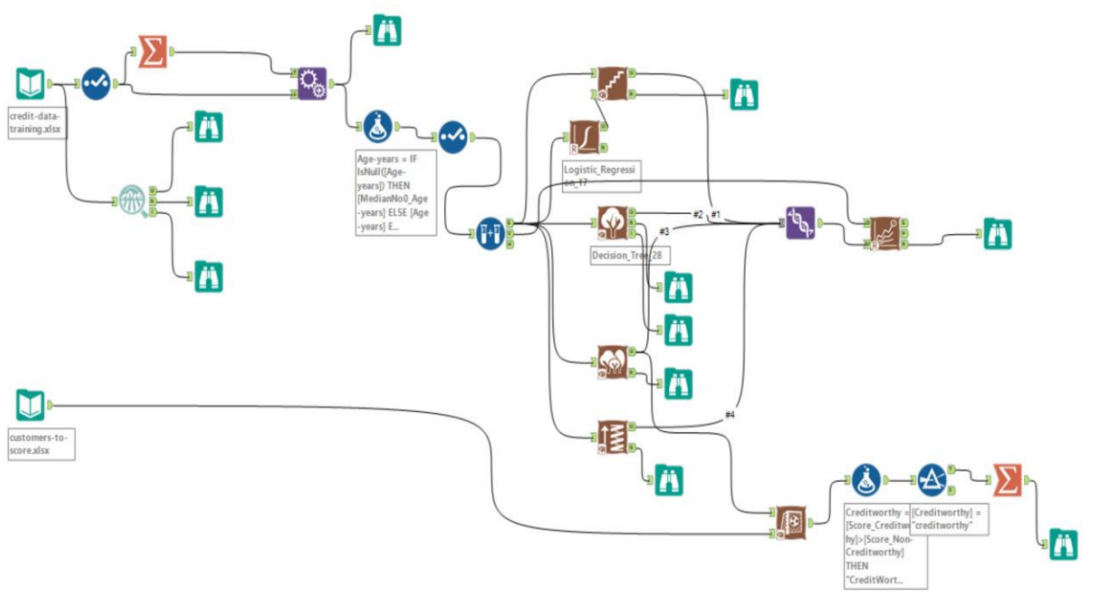
Logistic Regression Summary for Stepwise:The model has about a 76% accuracy but labels the creditworthy as 80% vs non creditworthy which is high therefore it is too biased towards creditworthiness.
For the summary of the Decision tree it was 67%
Boosted 78%
Forest Model 80%
In the Forest model we have three that are deemed non creditworthy but were actually creditworthy and 19 that 19 nonworthy that are actually non credit worthy. In the same vein 102 that were predicted to be creditworthy and were vs 26 that were predicted to be creditworthy but weren't.
You should have four sets of questions answered. (500 word limit)

Step 4: Writeup

Decide on the best model and score your new customers. For reviewing consistency, if Score_Creditworthy is greater than Score_NonCreditworthy, the person should be labeled as "Creditworthy"

Write a brief report on how you came up with your classification model and write down how many of the new customers would qualify for a loan. (250 word limit)

Some configurations I made:
Added and configured logistic regression model and stepwise tool
Added Decision Tree model
Added Boosted Model
Added Forest Model
Used Model Comparison Tool



Answer these questions:

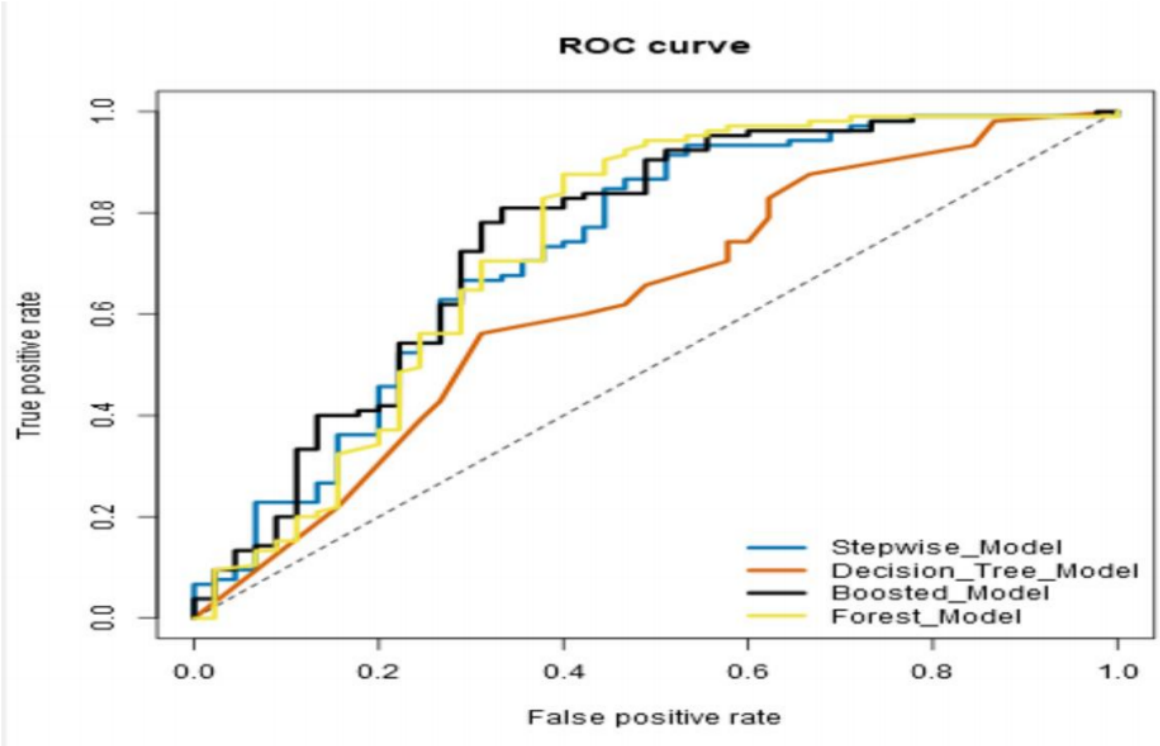
- Which model did you choose to use? Please justify your decision using **all** of the following techniques. Please only use these techniques to justify your decision:
    - Overall Accuracy against your Validation set

        I picked the Forest Model as that has the number at 0.8607 for non-creditworthiness which is the highest amongst the other models while the creditworthiness is still the second highest. That means it is quite high in both variables.
    - Accuracies within "Creditworthy" and "Non-Creditworthy" segments

Forest Model,as the accuracy is 0.7969. Although it is less than the stepwise model, it is a better model overall. And 86% for noncreditworthy.

- ROC graph

  The yellow line is the Forest Model



- Bias in the Confusion Matrices

Confusion matrix of Forest_Model

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	102	26
Predicted_Non-Creditworthy	3	19

Summary: The yellow line is at a positive rate the highest overall.Because this is a binary decision, we need accuracy on both sides which is why it is so important that our models are accurate for all sides. This is what makes it important to a lender.

Note: Remember that your boss only cares about prediction accuracy for Creditworthy and Non-Creditworthy segments.

- How many individuals are creditworthy?
  408 individuals

Before you Submit

Please check your answers against the requirements of the project dictated by the rubric here. Reviewers will use this rubric to grade your project.